Week 1 Session B

Thursday, Jan 15 2026

AI in Our Lives

- Where do you see it?
- What applications do you think use AI?

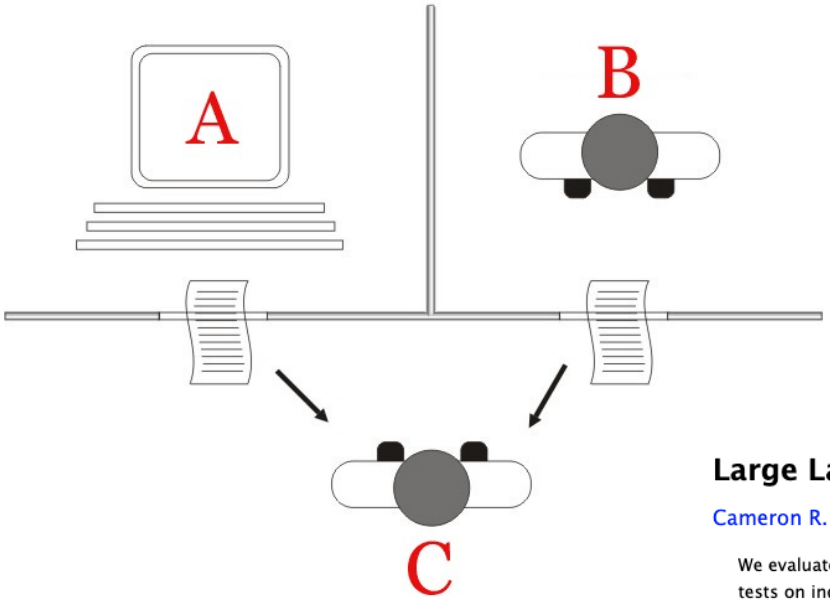
- Create a Google doc and we will work on it together

What is Artificial Intelligence? (I)

- What do people say?
  - BBC clip
  - Simplilearn
  - LearnFree
  - WhiteboardDoodles
  - From IBM
- Now let's judge them
  - When was the source published?
  - Who published them?
  - Are you seeing any contraindication? Who might be right?

What is Artificial Intelligence? (II)

- Turing Test “The Imitation Game” (1950)



Large Language Models Pass the Turing Test

Cameron R. Jones, Benjamin K. Bergen

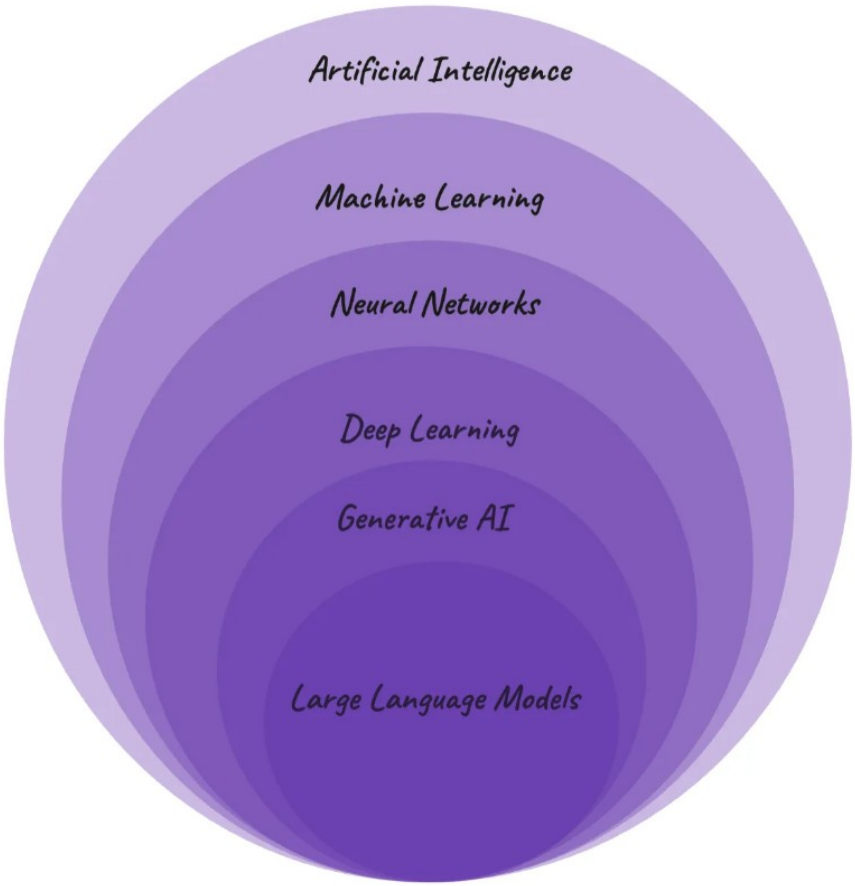
We evaluated 4 systems (ELIZA, GPT-4o, LLaMa-3.1-405B, and GPT-4.5) in two randomised, controlled, and pre-registered Turing tests on independent populations. Participants had 5 minute conversations simultaneously with another human participant and one of these systems before judging which conversational partner they thought was human. When prompted to adopt a humanlike persona, GPT-4.5 was judged to be the human 73% of the time: significantly more often than interrogators selected the real human participant. LLaMa-3.1, with the same prompt, was judged to be the human 56% of the time -- not significantly more or less often than the humans they were being compared to -- while baseline models (ELIZA and GPT-4o) achieved win rates significantly below chance (23% and 21% respectively). The results constitute the first empirical evidence that any artificial system passes a standard three-party Turing test. The results have implications for debates about what kind of intelligence is exhibited by Large Language Models (LLMs), and the social and economic impacts these systems are likely to have.

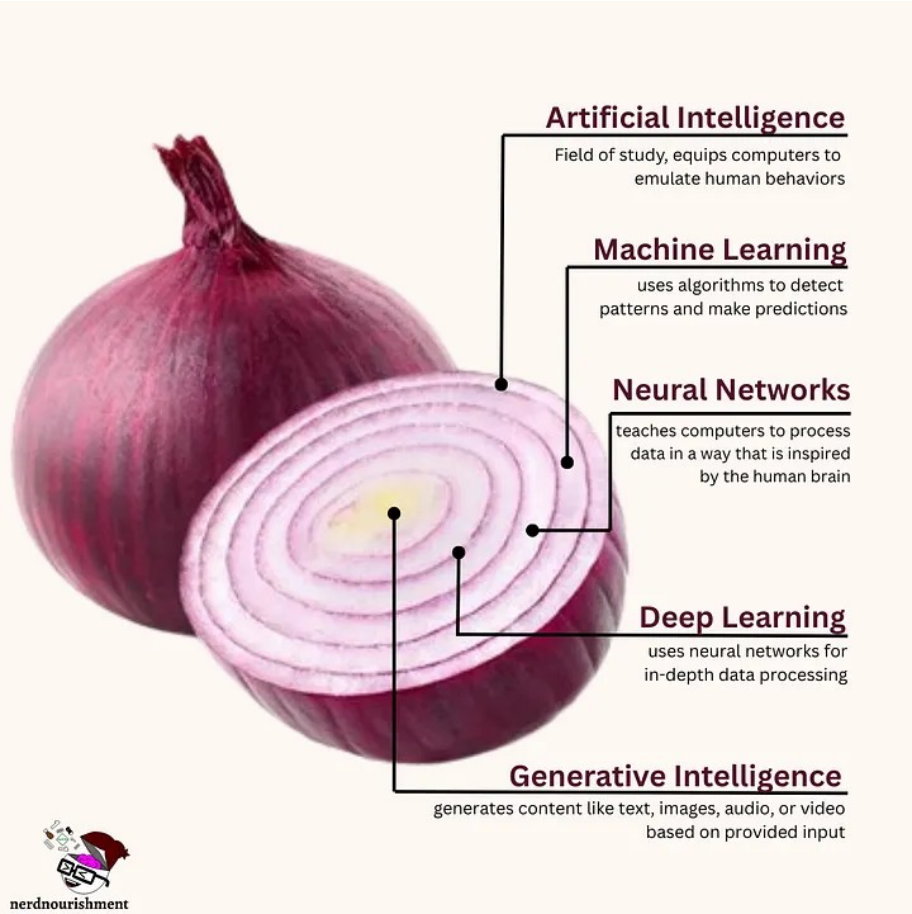
Subjects: **Computation and Language (cs.CL)**; Human-Computer Interaction (cs.HC)
Cite as: arXiv:2503.23674 **[cs.CL]**
(or arXiv:2503.23674v1 **[cs.CL]** for this version)
https://doi.org/10.48550/arXiv.2503.23674

What is Artificial Intelligence? (III)

- Applications:
  - Google AlphaGo vs Lee Sedol (2016): Go game
    - AlphaGo Movie 2017
  - Garry Kasparov vs IBM computer Deep Blue (1997): chess match

Look at the Onion (Hierarchy)

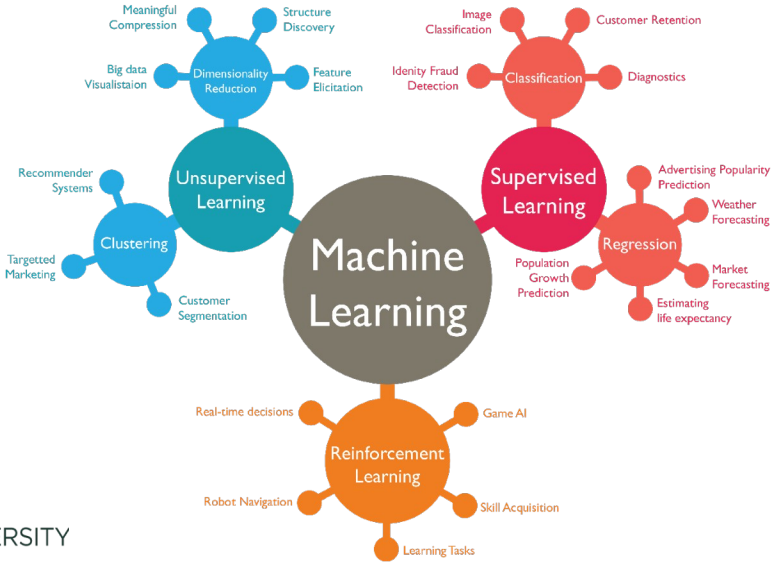




Machine Learning

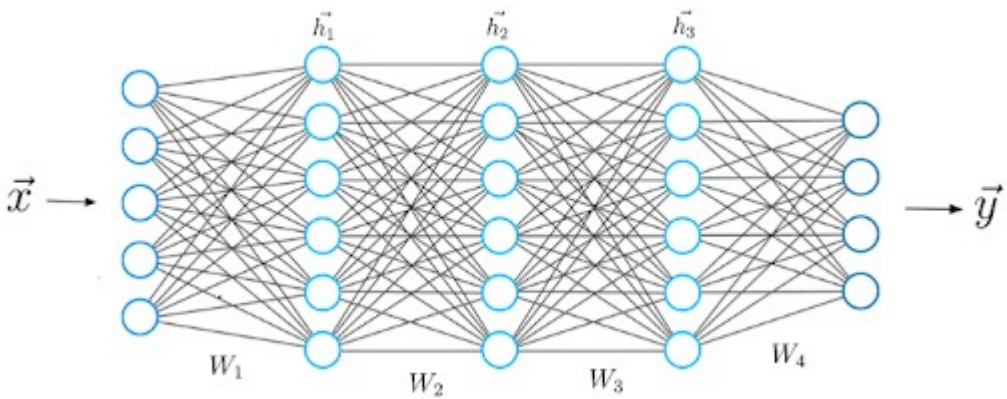
- Machine learning is a branch of AI.
- Machine Learning refers to the methods and algorithms that enable AI systems to identify patterns, make decisions, and improve themselves through all sources of input data (Sun et al.).

- Applications:
  - Siri, Alexa
  - Amazon's recommendation system
  - Speech language recognition
  - Spotify's daily mix



Neural Network

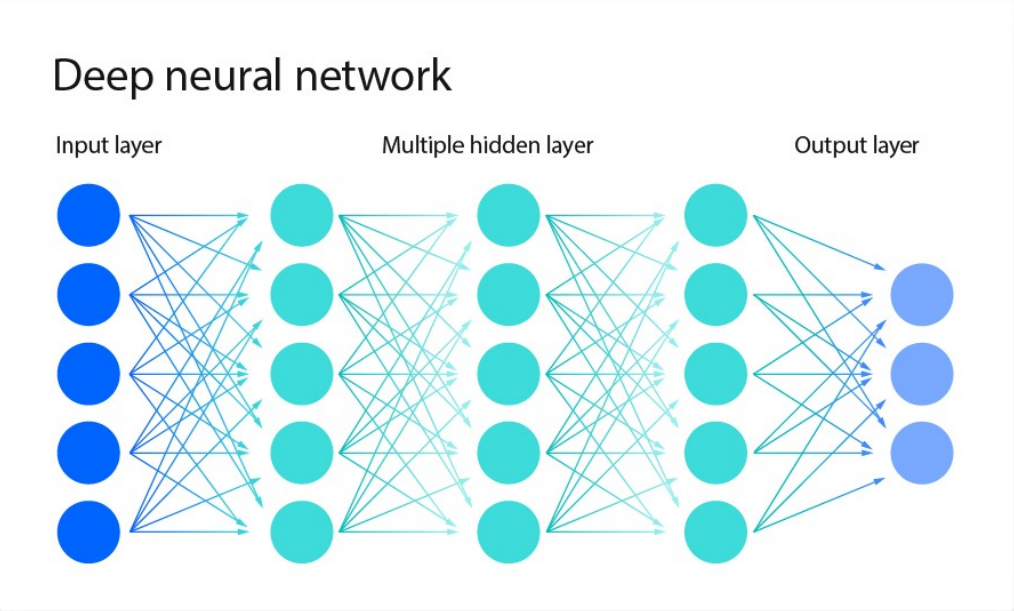
- Neural network is a subset of machine learning.
- Neural networks are inspired by the human brain, mimicking the way that biological neurons signal to one another (Caprasi).
- They are algorithms modeled after the brain and designed to recognize patterns.
- They work by taking an input and producing an output (prediction and decision).



- Applications/Examples:
  - Doing laundry
  - Your friend's handwriting
  - Machine being able to recognize cat/dog

Deep Learning

- Deep learning is a subset of machine learning driven by multilayered neural networks whose design is inspired by the structure of the human brain (IBM).
- Able to process more complicated task through huge amounts of data.



- Applications/Examples:
  - You being able to recognize your friend→ your phone has face recognition

AI vs Gen AI

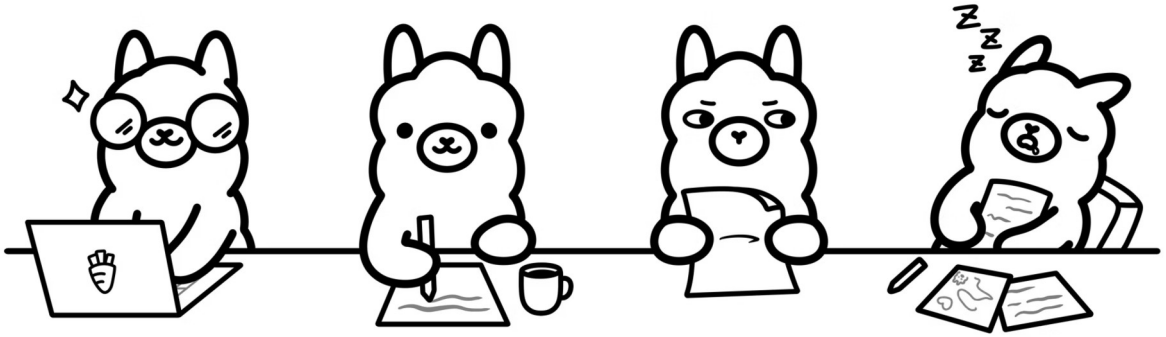
- AI is a discipline, it's a field of study.
- A common definition of AI is that it is a technology that enables machines to imitate various complex human skills (Sheikh et al.).

- Generative AI is a type of artificial intelligence technology that applies computational techniques, such as machine learning and deep learning, to generate seemingly new content in the forms of text, images, audio, and video from training data (Yu and Guo; Feuerriegel et al.).
- ChatGPT in 2022

Large Language Models (LLMs)

- A LLM is a form of gen AI. LLM focuses on generating human-like text based on the patterns learned from vast amounts of textual data during training process (Caprasi).

- Language Models: Gemini, Claude, GPT4/5, Deepseek, Llama, Gork, Qwen, Mistral, BERT …
- Who owns them?
- Where are they based?



Heroes we don’t talk about

- Ada Lovelace
- Grace Hopper

AI in Our Lives (II)

- Reflecting on what we have talked about today, would you like to make some improvements of your document?
  - Make updates in your Google doc

References

1. Caprasi, Chiara. “Artificial Intelligence, Machine Learning , Deep Learning & Generative AI and More.” Women in Technology, 24 July 2023, https://medium.com/womenintechnology/ai-c3412c5aa0ac.
2. Feuerriegel, Stefan, et al. “Generative AI.” Business & Information Systems Engineering, vol. 66, no. 1, Feb. 2024, pp. 111–26.
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5. Sun, Kai, et al. “Artificial Intelligence and Machine Learning: Definition of Terms and Current Concepts in Critical Care Research.” Journal of Critical Care, vol. 82, Aug. 2024, p. 154792.
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